

# 6. Orthogonality and Least Squares

## 6.1. Inner Product, Length, and Orthogonality

### Inner Product (Dot Product)

If  $\vec{u}$  and  $\vec{v}$  are vectors in  $\mathbb{R}^n$ , the number  $\vec{u} \cdot \vec{v}$  is called the inner product / dot product of  $\vec{u}$  and  $\vec{v}$ .

### Theorem 1

$$\begin{aligned} \text{a. } \vec{u} \cdot \vec{v} &= \vec{v} \cdot \vec{u} \\ \text{b. } \vec{u} \cdot (\vec{v} + \vec{w}) &= \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w} \\ \text{c. } (c\vec{u}) \cdot \vec{v} &= c(\vec{u} \cdot \vec{v}) = \vec{u} \cdot (c\vec{v}) \\ \text{d. } \vec{u} \cdot \vec{u} &\geq 0, \text{ and } \vec{u} \cdot \vec{u} = 0 \text{ if and only if } \vec{u} = \vec{0} \\ (c_1\vec{u}_1 \dots c_p\vec{u}_p) \cdot \vec{w} &= c_1(\vec{u}_1 \cdot \vec{w}) \dots c_p(\vec{u}_p \cdot \vec{w}) \end{aligned}$$

### Definition: Length/Norm

The length (or norm) of a vector  $\vec{v}$  in  $\mathbb{R}^n$  is the nonnegative scalar  $\|\vec{v}\|$ , defined by:

$$\|\vec{v}\| = \sqrt{\vec{v} \cdot \vec{v}} = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$$

and then

$$\begin{aligned} \|\vec{v}\|^2 &= \vec{v} \cdot \vec{v} \\ \|c\vec{v}\| &= |c|\|\vec{v}\| \end{aligned}$$

### Definition: Unit Vector and Normalize

A vector whose length is 1 is a unit vector. The process of creating a unit vector from  $\vec{v}$  is normalizing  $\vec{v}$ , and the unit vector is in the same direction as  $\vec{v}$ .

$$\vec{u} = \vec{v} / \|\vec{v}\|$$

### Definition: Distance

For  $\vec{u}$  and  $\vec{v}$  in  $\mathbb{R}^n$ , the distance between  $\vec{u}$  and  $\vec{v}$ , written as  $\text{dist}(\vec{u}, \vec{v})$ , is the length of the vector  $\vec{u} - \vec{v}$ :

$$\text{dist}(\vec{u}, \vec{v}) = \|\vec{u} - \vec{v}\|$$

### Definition: Orthogonal

Two vectors  $\vec{u}$  and  $\vec{v}$  in  $\mathbb{R}^n$  are said to be orthogonal (or perpendicular) if  $\vec{u} \cdot \vec{v} = 0$ .

### Theorem 2: The Pythagorean Theorem

Two vectors  $\vec{u}$  and  $\vec{v}$  are orthogonal if and only if:

$$\|\vec{u} + \vec{v}\|^2 = \|\vec{u}\|^2 + \|\vec{v}\|^2$$

# Orthogonal Complement

If a vector  $\vec{z}$  is orthogonal to every vector in a subspace  $W$  of  $\mathbb{R}^n$ , then  $\vec{z}$  is said to be orthogonal to  $W$ . The set of all such vectors  $\vec{z}$  is called the orthogonal complement of  $W$ , denoted by  $W^\perp$ :

$$W^\perp = \{\vec{z} \in \mathbb{R}^n \mid \vec{z} \cdot \vec{w} = 0 \text{ for all } \vec{w} \in W\}$$

## Properties of Orthogonal Complement

1.  $W^\perp$  is a subspace of  $\mathbb{R}^n$ .
2. a vector  $\vec{x}$  is in  $W^\perp$  if and only if  $\vec{x}$  is orthogonal to every vector in a set that spans  $W$ .

## Theorem 3

Let  $A$  be an  $m \times n$  matrix. The orthogonal complement of the row space of  $A$  is the null space of  $A$ , and the orthogonal complement of the column space of  $A$  is the null space of  $A^\top$ :

$$\begin{aligned}(\text{Row } A)^\perp &= \text{Null } A \\(\text{Col } A)^\perp &= \text{Null } A^\top\end{aligned}$$

## 6.2. Orthogonal Sets

### Definition: Orthogonal Set

A set of vectors  $\{\vec{u}_1, \dots, \vec{u}_p\}$  in  $\mathbb{R}^n$  is an orthogonal set if each pair of distinct vectors from the set is orthogonal. That is,  $\vec{u}_i \cdot \vec{u}_j = 0$  when  $i \neq j$ .

## Theorem 4

If  $S = \{\vec{u}_1, \dots, \vec{u}_p\}$  is an orthogonal set of nonzero vectors in  $\mathbb{R}^n$ , then  $S$  is linearly independent and hence is a basis for the subspace spanned by  $S$ .

### Definition: Orthogonal Basis

An orthogonal basis for a subspace  $W$  of  $\mathbb{R}^n$  is a basis for  $W$  that is also an orthogonal set.

## Theorem 5

Let  $\{\vec{u}_1, \dots, \vec{u}_p\}$  be an orthogonal basis for a subspace  $W$  of  $\mathbb{R}^n$ . For each  $\vec{y}$  in  $W$ , the weights in the linear combination  $\vec{y} = c_1\vec{u}_1 + c_2\vec{u}_2 + \dots + c_p\vec{u}_p$  are given by:

$$c_i = \frac{\vec{y} \cdot \vec{u}_i}{\vec{u}_i \cdot \vec{u}_i} \quad (i = 1, 2, \dots, p)$$

## Orthogonal Projection onto a Vector

Given a nonzero vector  $\vec{u}$  in  $\mathbb{R}^n$ , consider decomposing a vector  $\vec{y}$  in  $\mathbb{R}^n$  into a sum of two vectors: one a multiple of  $\vec{u}$ , and the other orthogonal to  $\vec{u}$ :

$$\vec{y} = \hat{\vec{y}} + \vec{z}$$

where  $\hat{\vec{y}} = \alpha\vec{u}$  (a scalar multiple of  $\vec{u}$ ) and  $\vec{z} = \vec{y} - \hat{\vec{y}}$  (orthogonal to  $\vec{u}$ ).

## Condition for Orthogonality

$\vec{z} = \vec{y} - \alpha\vec{u}$  is orthogonal to  $\vec{u}$  if and only if:

$$0 = (\vec{y} - \alpha\vec{u}) \cdot \vec{u} = \vec{y} \cdot \vec{u} - \alpha(\vec{u} \cdot \vec{u})$$

Solving for  $\alpha$  gives:

$$\alpha = \frac{\vec{y} \cdot \vec{u}}{\vec{u} \cdot \vec{u}}$$

## Definitions

- The vector  $\hat{\vec{y}} = \frac{\vec{y} \cdot \vec{u}}{\vec{u} \cdot \vec{u}} \vec{u}$  is the orthogonal projection of  $\vec{y}$  onto  $\vec{u}$ .
- The vector  $\vec{z} = \vec{y} - \hat{\vec{y}}$  is the component of  $\vec{y}$  orthogonal to  $\vec{u}$ .

Sometimes  $\hat{\vec{y}}$  is denoted by  $\text{proj}_L \vec{y}$  and is called the orthogonal projection of  $\vec{y}$  onto  $L$ . That is

$$\hat{\vec{y}} = \text{proj}_L \vec{y} = \frac{\vec{y} \cdot \vec{u}}{\vec{u} \cdot \vec{u}} \vec{u}$$

## Definition: Orthonormal Set

A set  $\{\vec{u}_1, \dots, \vec{u}_p\}$  is an orthonormal set if it is an orthogonal set of unit vectors (i.e.,  $\|\vec{u}_i\| = 1$  for all  $i$ , and  $\vec{u}_i \cdot \vec{u}_j = 0$  for  $i \neq j$ ). If  $W$  is the subspace spanned by such a set, then the set is an orthonormal basis for  $W$ .

## Theorem 6

An  $m \times n$  matrix  $U$  has orthonormal columns if and only if  $U^\top U = I_n$  (where  $I_n$  is the  $n \times n$  identity matrix).

## Theorem 7

Let  $U$  be an  $m \times n$  matrix with orthonormal columns, and let  $\vec{x}$  and  $\vec{y}$  be in  $\mathbb{R}^n$ . Then:

- a.  $\|U\vec{x}\| = \|\vec{x}\|$  (preserves length)
- b.  $(U\vec{x}) \cdot (U\vec{y}) = \vec{x} \cdot \vec{y}$  (preserves inner product)
- c.  $(U\vec{x}) \cdot (U\vec{y}) = 0$  if and only if  $\vec{x} \cdot \vec{y} = 0$  (preserves orthogonality)

## Definition: Orthogonal Matrix

An  $n \times n$  matrix  $U$  is an orthogonal matrix if its columns are orthonormal (equivalently,  $U^\top U = I_n$  and  $UU^\top = I_n$ ). This implies  $U^\top = U^{-1}$  (the transpose is the inverse).

## Properties of Orthogonal Matrices

1. If  $U$  is an orthogonal matrix, then  $U^{-1} = U^\top$ .
2.  $\det(U) = 1$  or  $\det(U) = -1$  (absolute value of determinant is 1).
3. if  $A$  and  $B$  are orthogonal matrix, then  $AB$  is an orthogonal matrix.

## Definition: Orthogonal Transformation

If  $U$  is an orthogonal matrix, then the linear transformation  $\vec{y} = U\vec{x}$  is an orthogonal transformation. (Preserves length and orthogonality, by Theorem 7)

## 6.3. Orthogonal Projection

### Theorem 8: The Orthogonal Decomposition Theorem

Let  $W$  be a subspace of  $\mathbb{R}^n$ . Then each  $\vec{y}$  in  $\mathbb{R}^n$  can be written uniquely in the form:

$$\vec{y} = \hat{\vec{y}} + \vec{z}$$

where  $\hat{\vec{y}} \in W$  and  $\vec{z} \in W^\perp$ . If  $\{\vec{u}_1, \dots, \vec{u}_p\}$  is an orthogonal basis of  $W$ , then:

$$\hat{\vec{y}} = \frac{\vec{y} \cdot \vec{u}_1}{\vec{u}_1 \cdot \vec{u}_1} \vec{u}_1 + \dots + \frac{\vec{y} \cdot \vec{u}_p}{\vec{u}_p \cdot \vec{u}_p} \vec{u}_p$$

and  $\vec{z} = \vec{y} - \hat{\vec{y}}$ .

### Definition: Orthogonal Projection onto a Subspace

The vector  $\hat{\vec{y}}$  in the orthogonal decomposition is called the orthogonal projection of  $\vec{y}$  onto  $W$ , denoted  $\text{proj}_W \vec{y}$ .

If  $\vec{y}$  is in  $W = \text{Span}\{\vec{u}_1, \dots, \vec{u}_p\}$ , then  $\text{proj}_W \vec{y} = \vec{y}$ .

### Theorem 9: The Best Approximation Theorem

Let  $W$  be a subspace of  $\mathbb{R}^n$ , let  $\vec{y}$  be any vector in  $\mathbb{R}^n$ , and let  $\hat{\vec{y}}$  be the orthogonal projection of  $\vec{y}$  onto  $W$ . Then  $\hat{\vec{y}}$  is the closest point in  $W$  to  $\vec{y}$ —that is:

$$\|\vec{y} - \hat{\vec{y}}\| < \|\vec{y} - \vec{w}\|$$

for all  $\vec{w} \in W$  with  $\vec{w} \neq \hat{\vec{y}}$ . The vector  $\hat{\vec{y}}$  is the best approximation to  $\vec{y}$  by elements of  $W$ .

## Projection Matrix

If  $W$  is a subspace of  $\mathbb{R}^n$  with orthogonal basis  $\{\vec{u}_1, \dots, \vec{u}_p\}$ , then the orthogonal projection of  $\vec{y}$  onto  $W$  can be written as  $\text{proj}_W \vec{y} = P\vec{y}$ , where  $P$  is the projection matrix:

$$P = \vec{u}_1 \vec{u}_1^\top + \vec{u}_2 \vec{u}_2^\top + \dots + \vec{u}_p \vec{u}_p^\top$$

If  $U = [\vec{u}_1 \quad \vec{u}_2 \quad \dots \quad \vec{u}_p]$  (an  $n \times p$  matrix with columns as the orthogonal basis), then:

$$P = U(U^\top U)^{-1}U^\top = UU^\top$$

(If the basis is orthonormal,  $U^\top U = I_p$ , so  $P = UU^\top$ )

## 6.4. The Gram-Schmidt Process

### Theorem 11: The Gram-Schmidt Process

Given a basis  $\{\vec{x}_1, \dots, \vec{x}_p\}$  for a nonzero subspace  $W$  of  $\mathbb{R}^n$ , define:

$$\begin{aligned}
\vec{v}_1 &= \vec{x}_1 \\
\vec{v}_2 &= \vec{x}_2 - \frac{\vec{x}_2 \cdot \vec{v}_1}{\vec{v}_1 \cdot \vec{v}_1} \vec{v}_1 \\
\vec{v}_3 &= \vec{x}_3 - \frac{\vec{x}_3 \cdot \vec{v}_1}{\vec{v}_1 \cdot \vec{v}_1} \vec{v}_1 - \frac{\vec{x}_3 \cdot \vec{v}_2}{\vec{v}_2 \cdot \vec{v}_2} \vec{v}_2 \\
&\vdots \\
\vec{v}_p &= \vec{x}_p - \frac{\vec{x}_p \cdot \vec{v}_1}{\vec{v}_1 \cdot \vec{v}_1} \vec{v}_1 - \frac{\vec{x}_p \cdot \vec{v}_2}{\vec{v}_2 \cdot \vec{v}_2} \vec{v}_2 - \cdots - \frac{\vec{x}_p \cdot \vec{v}_{p-1}}{\vec{v}_{p-1} \cdot \vec{v}_{p-1}} \vec{v}_{p-1}
\end{aligned}$$

Then  $\{\vec{v}_1, \dots, \vec{v}_p\}$  is an orthogonal basis for  $W$ , and  $\text{Span}\{\vec{v}_1, \dots, \vec{v}_k\} = \text{Span}\{\vec{x}_1, \dots, \vec{x}_k\}$  for each  $k = 1, 2, \dots, p$ .

## Theorem 12: The QR Factorization

If  $A$  is an  $m \times n$  matrix with linearly independent columns, then  $A$  can be factored as  $A = QR$ , where:

- $Q$  is an  $m \times n$  matrix whose columns form an orthonormal basis for  $\text{Col } A$ .
- $R$  is an  $n \times n$  upper triangular invertible matrix with positive entries on its diagonal.

### Relationships

$$\begin{aligned}
R &= Q^T A \\
Q^T Q &= I_n \\
R_{ii} &> 0 \quad (i = 1, 2, \dots, n)
\end{aligned}$$

## 6.5. Least-Squares Problems

### Definition: Least-Squares Solution

Given an  $m \times n$  matrix  $A$  and a vector  $\vec{b}$  in  $\mathbb{R}^m$ , a least-squares solution of  $A\vec{x} = \vec{b}$  is a vector  $\hat{\vec{x}}$  in  $\mathbb{R}^n$  such that:

$$\|A\hat{\vec{x}} - \vec{b}\| \leq \|A\vec{x} - \vec{b}\|$$

for all  $\vec{x}$  in  $\mathbb{R}^n$ . The vector  $A\hat{\vec{x}}$  is the orthogonal projection of  $\vec{b}$  onto  $\text{Col } A$ , and  $\|A\hat{\vec{x}} - \vec{b}\|$  is the least-squares error.

### Theorem 13

The set of least-squares solutions of  $A\vec{x} = \vec{b}$  coincides with the nonempty set of solutions of the normal equations:

$$A^T A \vec{x} = A^T \vec{b}$$

### Theorem 14

Let  $A$  be an  $m \times n$  matrix. The following are equivalent:

- The equation  $A\vec{x} = \vec{b}$  has a unique least-squares solution for each  $\vec{b}$  in  $\mathbb{R}^m$ .
- The columns of  $A$  are linearly independent.
- The matrix  $A^T A$  is invertible.

When these conditions hold, the unique least-squares solution is:

$$\hat{\vec{x}} = (A^T A)^{-1} A^T \vec{b}$$

The distance from  $\vec{b}$  to  $A\vec{x}$  is the least-squares error of this approximation.

## Theorem 15

Given an  $m \times n$  matrix  $A$  with linearly independent columns, let  $A = QR$  be a QR factorization of  $A$  (as in Theorem 12). Then for each  $\vec{b}$  in  $\mathbb{R}^m$ , the equation  $A\vec{x} = \vec{b}$  has a unique least-squares solution given by:

$$\hat{\vec{x}} = R^{-1}Q^T\vec{b}$$

### Proof

Let  $\hat{\vec{x}} = R^{-1}Q^T\vec{b}$ . Since  $A = QR$ , we have  $A\hat{\vec{x}} = QRR^{-1}Q^T\vec{b} = QQ^T\vec{b}$ . By Theorem 12, the columns of  $Q$  form an orthonormal basis for  $\text{Col } A$ , so  $QQ^T\vec{b}$  is the orthogonal projection of  $\vec{b}$  onto  $\text{Col } A$ . Thus,  $A\hat{\vec{x}} = \text{proj}_{\text{Col } A}\vec{b}$ , which shows that  $\hat{\vec{x}}$  is a least-squares solution. Uniqueness follows from Theorem 14.

## 6.7. Inner Product Spaces

### Definition: Inner Product on a Vector Space

An inner product on a vector space  $V$  is a function that, to each pair of vectors  $\vec{u}$  and  $\vec{v}$  in  $V$ , associates a real number  $\langle \vec{u}, \vec{v} \rangle$  and satisfies the following axioms for all  $\vec{u}, \vec{v}, \vec{w}$  in  $V$  and all scalars  $c$ :

1.  $\langle \vec{u}, \vec{v} \rangle = \langle \vec{v}, \vec{u} \rangle$  (symmetry)
2.  $\langle \vec{u} + \vec{v}, \vec{w} \rangle = \langle \vec{u}, \vec{w} \rangle + \langle \vec{v}, \vec{w} \rangle$  (additivity)
3.  $\langle c\vec{u}, \vec{v} \rangle = c\langle \vec{u}, \vec{v} \rangle$  (homogeneity)
4.  $\langle \vec{u}, \vec{u} \rangle \geq 0$ , and  $\langle \vec{u}, \vec{u} \rangle = 0$  if and only if  $\vec{u} = \vec{0}$  (positive definiteness)

A vector space with an inner product is an inner product space.

### Theorem 16: The Cauchy-Schwarz Inequality

For all  $\vec{u}, \vec{v}$  in an inner product space  $V$ :

$$|\langle \vec{u}, \vec{v} \rangle| \leq \|\vec{u}\| \|\vec{v}\|$$

where  $\|\vec{u}\| = \sqrt{\langle \vec{u}, \vec{u} \rangle}$  (norm induced by the inner product).

### Theorem 17: The Triangle Inequality

For all  $\vec{u}, \vec{v}$  in an inner product space  $V$ :

$$\|\vec{u} + \vec{v}\| \leq \|\vec{u}\| + \|\vec{v}\|$$

### Generalized Orthogonalization

Given a basis  $\{\vec{x}_1 \cdots \vec{x}_p\}$  for a nonzero subspace  $W$  of  $\mathbb{R}^n$ , define:

$$\begin{aligned} \vec{v}_1 &= \vec{x}_1 \\ \vec{v}_2 &= \vec{x}_2 - \frac{\langle \vec{x}_2, \vec{v}_1 \rangle}{\langle \vec{v}_1, \vec{v}_1 \rangle} \vec{v}_1 \\ &\vdots \\ \vec{v}_p &= \vec{x}_p - \sum_{i=1}^{p-1} \frac{\langle \vec{x}_p, \vec{v}_i \rangle}{\langle \vec{v}_i, \vec{v}_i \rangle} \vec{v}_i \end{aligned}$$

Then  $\{\vec{v}_1 \cdots \vec{v}_p\}$  is an orthogonal basis for  $W$ .

In addition,  $\text{Span}\{\vec{x}_1 \cdots \vec{x}_p\} = \text{Span}\{\vec{v}_1 \cdots \vec{v}_p\}$ .